HONEYBEE (Apis indica) - QUICK REVISION SHEET

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1. Introduction & Scientific Classification

- **Study of Honeybees:** Mellitology or Apiology
- **Scientific Name:** *Apis indica* (most common species)
- **Rearing / Agriculture:** Apiculture
- **Social Nature:** Highly social insects living in well-developed colonies/hives with distinct division of labor.

Taxonomic Rank	Classification
Kingdom	Animalia
Phylum	Arthropoda (jointed appendages)
Class	Insecta (three-part body, six legs)
Genus	*Apis*

2. External Structure & Morphology

Overall Size: 9 mm to 12 mm

A. Head

- **Compound Eyes:** 1 pair located at the top of the head; contains ~6,000 facets (ommatidia) for image formation.
- **Ocelli (Simple Eyes):** 3 ocelli positioned on the head to perceive light intensity.
- **Antennae:** 1 pair for environmental perception (sensory organs for touch and smell).

- **Mouthparts:** Adapted for chewing (mandibles) and a specialized **proboscis** for sucking nectar.

B. Thorax

- **Prothorax (Anterior):** Bears 1st pair of legs (forelegs).
- **Mesothorax (Middle):** Bears 2nd pair of legs (middle legs) and 1st pair of wings (forewings).
- **Metathorax (Posterior):** Bears 3rd pair of legs (hind legs) and 2nd pair of wings (hindwings).

C. Abdomen

- Consists of 9 segments normally (visible segments: **7 in males**, **6 in females**).
- In **worker bees**, the hindmost 3 segments are modified into a **stinger** (ovipositor modified for defense, rendering them incapable of laying eggs).

3. Hive Organization & Castes Comparison

Feature	Queen Bee	Drone Bee	Worker Bee
Sex & Fertility	Fertile Female	Fertile Male	Sterile Female
Population / Hive	1 per hive	100 – 500	20,000 – 80,000
Development Time	16 days	24 days	21 days
Origin / Genetics	Fertilized egg (fed Royal Jelly continuously)	Unfertilized egg (Parthenogenesis, 16 chromosomes)	Fertilized egg (fed Bee Bread / Honey after 3 days)
Primary Roles & Duties	• Lays eggs (1,500–3,000 per day) after nuptial flight • Produces hive-identifying pheromones	• Mates with the queen during nuptial flight • Helps keep the hive warm • *Note: Dies after mating due to*	• Nurtures eggs & larvae • Feeds queen Royal Jelly • Collects nectar & pollen • Cleans, constructs,

Feature	Queen Bee	Drone Bee	Worker Bee
	• Maintains hive temperature	*genital rupture; lacks foraging mouthparts/wings to collect honey.*	& defends hive

4. Life Cycle & Metamorphosis

Honeybees undergo **complete metamorphosis** consisting of four developmental stages: **Egg** → **Larva** → **Pupa** → **Adult**.

Stage Durations Summary Table

Caste	Egg Stage	Larval Stage	Pupa Stage	Total Time to Adult
Queen	3 days	5.5 days	7.5 days	**16 days**
Worker	3 days	6.0 days	12.0 days	**21 days**
Drone	3 days	7.0 days	14.0 days	**24 days**

Stage Details

1. **Egg Stage:** Size 1 mm – 1.5 mm. Laid by queen after mating. Hatches in 3 days.
    - *Fertilized eggs* → Develop into Queens or Workers.
    - *Unfertilized eggs* → Develop into Drones via **Parthenogenesis**.
2. **Larval Stage:** Rice-grain shaped with mouthparts. Termed a *"voracious eater"* due to massive food intake.
    - All larvae are fed **Royal Jelly** for the first 3 days.
    - If fed Royal Jelly continuously → Becomes **Queen**.
    - If switched to Honey & Bee Bread → Becomes **Worker** or **Drone**.
3. **Pupa Stage:** Non-feeding stage enclosed in a wax cell. Organ development occurs via

Histogenesis (formation of wings, eyes, legs). Adult chews through the wax seal to emerge.
4. **Adult Stage:** Fully developed bees emerge and assume specific hive roles.

5. Important Biological Terminology

- **Moulting:** Shedding/changing of skin during the larval stage (occurs ~4 times).
- **Histogenesis:** Process of tissue and organ differentiation (head, wings, mouthparts) during the pupal stage.
- **Metamorphosis:** Complete biological development process from Egg → Larva → Pupa → Adult.
- **Parthenogenesis:** Asexual reproduction where an unfertilized egg (haploid, 16 chromosomes) develops into a male drone without sperm fertilization.
- **Nuptial Flight ("Honeymoon Flight"):** High-altitude flight where the young queen mates mid-air with multiple drones. Occurs 1–2 times in her lifespan.
- **Spermatheca:** Specialized sac-like structure in the queen's reproductive system that stores sperm collected during the nuptial flight.
- **Swarming:** Departure of the old queen along with a crowd of worker bees to establish a new colony when a new queen emerges.
- **Foraging / Waggle Dance:** Communication dance patterns performed by worker bees to signal the direction and distance of food sources to other hive members.

6. Economic Products & Applications of Apiculture

Product / Role	Description & Value
1. Honey	Primary commercial product. Rich in carbohydrates, highly nutritious, and possesses antibacterial/medicinal properties.
2. Bee Venom (Melittin)	Collected from worker bee stingers. Used in high-grade pharmaceutical drugs for treating paralysis and cancer research. Estimated value: ~₹70 Lakhs / Litre.

Product / Role	Description & Value
3. Royal Jelly	Nutritious secretion produced by worker bees to feed queen larvae. Highly valued health supplement. Estimated value: ~₹70,000 – ₹90,000 / kg.
4. Beeswax	Secreted by worker bees to construct comb cells. Leftover after honey extraction; widely used in high-grade cosmetics, polishes, and candles.
5. Bee Propolis	Dark resinous glue collected from plants/glands used to seal hive gaps. Possesses antimicrobial properties used in medicines.
6. Vital Pollinators	Crucial ecological role: Honeybees carry out roughly **90% of cross-pollination** in flowering plants and agricultural crops.